

configuration) to switch the iron on via relay RL1. Here it gets heated for one second in every six seconds. This duty cycle is decided by R5, R6 and C3 when the iron is in the stand. The presence of the iron in its stand is sensed by electrical contacts P1 and P2.

When the iron is at rest, the base of T1 is pulled low and therefore T1 doesn't conduct. Conversely, when the iron is out of its stand, T1 conducts to pull pins 2 and 6 of IC2 high, inhibiting its operation. During this time, pin 3 of IC2 is low and therefore the iron is continuously powered via normally-closed (N/C) contact of relay RL1.

Construction and testing

An actual-size, single-side PCB for the soldering iron tip preserver is shown in Fig. 2 and its component layout in Fig. 3. Assemble the circuit on a PCB to minimise time and assembly errors.

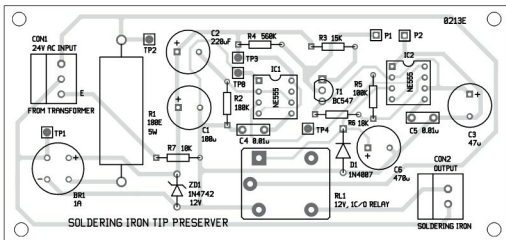


Fig. 3: Component layout for the PCB

Carefully assemble the components and check for any overlooked error. Suitable connectors are available on the PCB to connect the secondary winding of the transformer and the soldering iron. Some slight modification of the stand may be required to sense the presence of iron in the stand. The warm-up time and standby temperature can be varied by changing the

values of resistors R2 and R5.

To test the circuit for proper functioning, check test points TP1 and TP2 for correct power supply to the soldering iron. These should be at around 24V. Check TP3 for power supply to timers as mentioned in the table. The output of IC1 can be checked at TP4. ●

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